ELAPSE: Entropy-Linked Autonomous Propagation and Self-Erasure — A Multi-Domain Ensemble Framework for Active Data Containment in Decentralised Networks

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Abstract

Nonlinear diffusion of information across stochastic, decentralised peer-to-peer networks exhibits rich bifurcation behaviour: the spatial entropy $H(t)$ of the spreading payload undergoes a sharp qualitative transition near a critical threshold H_c, separating a subcritical phase (localised spreading, recoverable mass) from a supercritical phase (irreversible global diffusion). Chronological containment protocols [1, 2] are structurally blind to this bifurcation, firing on a fixed clock regardless of whether the dynamical system is in the subcritical or supercritical regime.

We introduce **ELAPSE** (Entropy-Linked Autonomous Propagation and Self-Erasure), a framework governed by the entropy itself as a bifurcation parameter. Five nonlinear mechanisms — diffusion, epidemic, stochastic-finance, biological, and cascade — each cast a continuous vote $v_k(t) \in [0, 1]$ measuring proximity to the critical threshold. These votes are fused by an entropy-regularised ensemble that minimises the *Time-Integrated Entropy*

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Exposure (IEE $= \int_0^T H(t)M(t)\,dt$ [nats $\times$ data-mass $\times$ time]), a functional that encodes both the geometric complexity of the spreading trajectory and the surviving data mass.

We establish five analytical results characterising the nonlinear dynamics. *(1) Well-posedness*: Pathwise-unique non-negative strong solutions via Yamada–Watanabe with CIR-type noise ($\sigma_n\sqrt{x_i}$). *(2) Spectral IEE bound*: IEE $\leq \ln(n) \cdot M_0 \cdot \tau^* + e^{-\mu\delta_{\min}(T-\tau^*)}\mathrm{IEE}_0$, with spectral trigger time $\tau^* = \frac{1}{2\alpha\lambda_2}\ln\!\left(\frac{C_0}{H_{\max}-H_c}\right)$. *(3) Bifurcation at H_c^**: The critical threshold $H_c^* = H_{\max} - C_0 e^{-2\alpha\lambda_2 T}$ separates subcritical (ELAPSE achieves bounded IEE) from supercritical (IEE grows without bound) regimes; transition width scales as $(2\alpha\lambda_2 T)^{-1}$. *(4) IEE landscape complexity*: The linearised IEE is exactly convex (Laplace-transform proof), but the full nonlinear SDE landscape exhibits a 14.1% Jensen-violation rate — a quantitative signature of the nonlinear IEE geometry induced by the bifurcation near H_c. *(5) Privacy guarantee*: IEE control implies $\mathbb{E}[\text{recovered mass}] \leq Qe^{-\mu\delta_{\min}T}/n$ for any Q-node adversary.

Numerical experiments across Erdős-Rényi, Barabási-Albert, and Watts-Strogatz topologies ($n \in \{50, 100, 200, 500\}$) confirm the bifurcation structure: ELAPSE suppresses IEE by at least $9\times$ relative to the uncontrolled baseline at $n = 500$, with the ensemble outperforming any single mechanism on BA and WS topologies. Real-world validation on Gnutella P2P networks confirms that the subcritical–supercritical transition is observable in empirical data.

Keywords: Nonlinear dynamics, Bifurcation, Complex networks, Shannon entropy, Self-erasure, Stochastic differential equations, IEE landscape, Phase transition, Chaos, Active data containment

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1. Introduction

1.1. Motivation and Problem Statement

The proliferation of sensitive data across decentralised peer-to-peer (P2P) networks constitutes a qualitatively different privacy threat than the storage of data in centralised repositories. Once a data payload escapes its point of origin — whether via legitimate forwarding, an insider disclosure, or endpoint compromise — the originating system loses all physical control over its further replication. The challenge is acute in the context of the European Union's *right to erasure* (GDPR Article 17 [3]), which mandates that data subjects can demand removal of their personal data; a mandate that is straightforwardly unenforceable once a payload has diffused across thousands of pseudonymous P2P nodes.

The scale of the problem is growing. IBM's 2024 Cost of a Data Breach Report [4] documented a global average breach cost of USD 4.88 million, a 10% year-on-year increase, with P2P exfiltration cited among the leading attack vectors. The 2024 Verizon Data Breach Investigations Report [5] further notes that data discovered in unauthorised copies on external networks has a median "time to identification" exceeding 280 days, during which the privacy exposure accumulates continuously.

1.2. Limitations of Chronological Containment

The existing active-containment literature — most influentially the Vanish [1] and FADE [2] systems — relies on a shared architectural assumption: a *chronological decay timer* governs data lifetime. Vanish overwrites Distributed Hash Table (DHT) shards on a fixed schedule (typically 8 hours), so that the decryption key is irretrievably lost after a wall-clock timeout. FADE extends this with file-assured deletion via policy-based key management.

The fundamental limitation of this paradigm is its independence from the actual network topology. On a scale-free (Barabási-Albert) graph, the hub-driven spreading mechanism can saturate a majority of nodes within a small fraction of any fixed timeout [6, 7]. A data payload that has already reached 10^3 nodes before the 8-hour timer fires may be practically irrecoverable regardless of key deletion. Conversely, on a sparse small-world (Watts-Strogatz) graph, the same timer may fire when the data has barely left the source neighbourhood, over-triggering deletion.

1.3. Entropy as a Bifurcation Parameter

The key insight of ELAPSE is that the network's spreading dynamics exhibit a genuine bifurcation: the qualitative behaviour of the system changes fundamentally near a critical entropy threshold H_c. We use the Shannon entropy of the data's spatial distribution as the bifurcation parameter:

$$H(t) = -\sum_{i=1}^{n} p_i(t) \ln p_i(t), \quad p_i(t) = \frac{x_i(t)}{\sum_j x_j(t)}, \tag{1}$$

where $x_i(t) \geq 0$ is the data concentration at node i at time t. When $H(t) < H_c$, the spreading is in the *subcritical phase*: data is concentrated on few nodes, deletion can arrest diffusion, and IEE remains bounded. When $H(t) \geq H_c$, the system enters the *supercritical phase*: data has spread broadly enough that erasure cannot prevent unbounded IEE accumulation. The boundary $H_c^* = H_{\max} - C_0 e^{-2\alpha\lambda_2 T}$ (Section 4.4) is a sharp phase-transition point; the transition width scales as $(2\alpha\lambda_2 T)^{-1}$, sharpening with time and algebraic connectivity.

This replaces the chronological clock with an *entropy-bifurcation clock*: self-erasure fires when the system is about to cross from subcritical to supercritical, the only moment at which deletion is both necessary and sufficient to contain the privacy damage.

1.4. Contributions

This paper makes the following contributions:

1. **Framework**: We introduce ELAPSE, fusing five mechanism-specific deletion votes via an entropy-regularised Nelder-Mead ensemble optimising IEE (Section 3–4).
2. **Rigorous theory** (Section 4.3): Five results: (i) pathwise-unique non-negative strong solution (Theorem 1, via Yamada-–Watanabe); (ii) a closed-form spectral IEE bound in terms of n, M_0, and deletion intensity (Theorem 2); (iii) a *topology-refined* IEE bound incorporating the algebraic connectivity λ_2, giving the spectral trigger time $\tau(\lambda_2)$ explicitly (Corollary 1); (iv) a residual-mass privacy guarantee connecting IEE to an exponential data-recovery bound for bounded adversaries (Proposition 1); (v) *exact* convexity of the linearised IEE objective (Theorem 3, Corollary 2); the full nonlinear SDE exhibits a 14.1% Jensen-violation rate, a direct quantitative signature of the nonlinear IEE landscape complexity induced by bifurcation near H_c.
3. **Statistical rigour**: All IEE values are reported as mean $\pm$ 95% CI over $N_{\text{test}} = 20$ out-of-sample seeds, three topologies, and four network sizes (Section 5).
4. **Real-world validation**: The full ensemble is evaluated on subgraphs of the Stanford SNAP Gnutella P2P datasets [8] (Section 6).

5. **Adversarial analysis**: We model an attacker who throttles propagation and propose an adaptive-H_c countermeasure (Section 7).
6. **Distributed estimation**: We propose a gossip-protocol approximation to the global entropy observable and quantify its trigger-accuracy (Section Appendix A).

1.5. Paper Organisation

Section 2 reviews the literature and positions ELAPSE. Section 3 defines the five core mechanisms. Section 4 presents the ensemble construction and five theoretical results (Theorems 1–3, Corollaries 1–2, Propositions 1–2), including the bifurcation analysis at H_c^* (Section 4.4) and the IEE landscape complexity characterisation (Section 4.5). Section 5 provides the main statistical evaluation. Section 6 reports real-world validation. Section 7 analyses adversarial evasion. Section Appendix C concludes. Appendix Appendix A covers distributed entropy estimation (gossip protocol); Appendix Appendix B provides a mechanistic analysis and fix for M3's underperformance on scale-free topologies.

2. Related Work

2.1. Information Diffusion, Entropy, and Nonlinear Dynamics on Graphs

The quantitative study of information propagation on complex networks has a rich lineage in nonlinear dynamics. Foundational epidemiological models for spreading processes [7] established that network topology — and especially heterogeneous degree distributions — fundamentally alters threshold phenomena, producing sharp bifurcations in spreading behaviour at critical parameter values. Universal resilience patterns [9] characterise tipping points and hysteresis under perturbation, confirming that complex networks generically exhibit bifurcation structure.

The particular role of Shannon entropy as a topological indicator was established by Zhao et al. [10], who demonstrated that multi-scale entropy metrics provide macroscopic proxies for information volatility across heterogeneous degree distributions. *ELAPSE departs from this line of work in a fundamental respect*: Zhao et al. use entropy as a passive *diagnostic* of spreading state, whereas ELAPSE uses $H(t)$ as an active *bifurcation parameter* that governs the system's own self-erasure dynamics. This transforms entropy from an observation tool into a control variable whose critical level H_c directly determines the subcritical–supercritical phase boundary. The resulting SDE system — with entropy-dependent mortality and CIR-type noise — is substantially nonlinear, and its IEE functional landscape exhibits genuine complexity (14.1% Jensen violation) absent from the passive monitoring framework of [10].

Parallel advances in continuous graph diffusion, originating in spectral graph signal processing [11] and extended through neural PDE-on-graph formulations (GRAND [12]; Eliasof et al. [13]; Fan et al. [14]), confirm that partial differential equations on graph topologies accurately capture non-linear propagation boundaries.

2.2. Static and Active Data Containment

Early cryptographic containment schemes were pioneered by Vanish [1], which exploited DHT key churn to implement passive key decay. Wolchok et al. [15] demonstrated Sybil-attack vulnerability. FADE [2] and the Ephemerizer [16] strengthened storage layers but retained chronological timers. Rondeau and Temple [17] survey secure deletion approaches and confirm that time-blind mechanisms are structurally insufficient for dynamic network environments. Truex et al. [18] demonstrate a hybrid local/global approach to privacy-preserving federated learning in distributed settings, showing that no single privacy mechanism dominates across heterogeneous network topologies — a finding consistent with our ensemble design.

2.3. Machine Unlearning and Algorithmic Forgetting

The machine-learning community has recently formalised the complementary problem of "algorithmic forgetting": ensuring a trained model provides no residual information about a removed training sample. Bourtoule et al. [19] established the SISA training framework, Nguyen et al. [20] and Xu et al. [21] provide comprehensive surveys of unlearning algorithms, highlighting the need for topological removal verification. Oesterling et al. [22] incorporate fairness constraints into active data removal, which motivates our entropy-regularised ensemble approach.

2.4. Differential Privacy and Information-Theoretic Bounds

Dwork and Roth's foundational treatise [23] establishes differential privacy as the canonical framework for information-theoretic privacy guarantees. While differential privacy bounds *inference leakage* from a fixed dataset, ELAPSE addresses a complementary problem: minimising the *spatial exposure* of a diffusing payload before deletion can take effect. These two regimes are formally distinct but share the use of entropy-based metrics as core sensitivity measures.

2.5. Control of Diffusion Processes on Graphs

ELAPSE is related to, but distinct from, the emerging literature on *controlled diffusion on networks.* Mean-field game (MFG) frameworks [24, 25] study populations of McKean-–Vlasov-type agents interacting through their empirical distribution, deriving feedback control laws from coupled Hamilton-–Jacobi-–Bellman and Fokker-–Planck equations. In MFG language, ELAPSE is a *social planner* problem: we seek a mortality schedule $\mu\Delta(t; \mathbf{w})$ that minimises a population-level cost (IEE) subject to the coupled McKean-–Vlasov SDE (**??**). However, ELAPSE differs from classical MFG in three ways: (i) topology enters via the Laplacian L rather than a mean-field coupling term; (ii) the control $\Delta(t; \mathbf{w})$ is an entropy-dependent functional of the empirical distribution; and (iii) we optimise over a finite ensemble of pre-specified mechanisms rather than solving a PDE. Online feedback control on network flow manifolds [26] addresses a related setting but focuses on power systems rather than information-theoretic objectives. ELAPSE contributes to this space with the first entropy-functional IEE bound (Theorem 2) and spectral trigger time (Corollary 1) for controlled diffusion on heterogeneous P2P graphs.

2.6. Positioning of ELAPSE

Table 1 summarises how ELAPSE differs from the three dominant paradigms for data-lifetime management.

ELAPSE occupies a unique position in the landscape: it is the first framework that (i) replaces the chronological erasure clock with a spatial entropy trigger, (ii) constructs a regularised ensemble across five physical domains,

Table 1: Comparison of data-deletion paradigms. ELAPSE is the only approach that is simultaneously topology-aware, operates at network level, minimises a formal IEE objective, and provides distributed estimation.

Paradigm	Topology-aware	Network-level	Formal IEE obj.	Distributed estimation
Cryptographic (Vanish, FADE) [1, 2]	✗	✗	✗	✗
Chron. timer (M7, §5.8)	✗	✗	✗	✗
Machine unlearning [19, 21]	✗	✗	✗	✗
Differential privacy [23]	✗	✗	✗	Partial
ELAPSE (ours)	✓	✓	✓	✓

(iii) provides rigorous theoretical guarantees (Theorems 1–3) backed by statistical evaluation, and (iv) characterises the minimum gossip hop count k^* as a function of network algebraic connectivity (Proposition 3). Prior ensemble approaches for network spreading (e.g., competing epidemics [27]; social cascades [28]) do not address active containment.

3. The Five Core Mechanisms

3.1. Network and Notation

Let $G = (V, E)$ be an undirected connected graph with $|V| = n$ nodes. The graph Laplacian $L = D - A$ (where D is the degree matrix and A the adjacency matrix) and the Fiedler value λ_2 (second-smallest eigenvalue of L, the algebraic connectivity [29]) play central roles. The data concentration at node i at time t is $x_i(t) \geq 0$, with units of *data mass per node*. Initial conditions are $x_i(0) = u_i$ where $u_i \sim \text{Uniform}(0.5, 1.0)$ for a randomly chosen 10% of nodes (the initial data sources) and $x_i(0) = 0$ otherwise; total initial mass $M(0) = \sum_i x_i(0) \in (0, n/10]$. All IEE values therefore have units of [nats $\times$ data-mass $\times$ time], and scale with both n and $M(0)$. Spatial privacy exposure is measured by $H(t)$ as defined in Eq. (1), with $H_{\max} = \ln n$ and a critical threshold $H_c = 0.65 H_{\max}$ used throughout.

Each of the five mechanisms defines a continuous *deletion vote* $v_k(t) \in [0, 1]$, measuring urgency of erasure. The composite mortality pressure $\Delta(t) = \sum_k w_k v_k(t)$ drives data removal.

3.2. M1: Entropy-Gated Diffusion Model (EGDM)

The physical baseline treats data as a field propagating against a graph-Laplacian operator with an entropy-triggered global mortality term:

$$\frac{d\mathbf{x}}{dt} = -\alpha L\mathbf{x} - \mu\, \sigma(H(t))\, \mathbf{x} + \mathbf{s}(t) + dW_t, \tag{2}$$

where $\alpha = 0.3$ is the diffusion coefficient, $\mu = 1.5$ the mortality rate, $\mathbf{s}(t)$ a small external injection, and dW_t a multiplicative CIR-type Wiener noise with scale $\sigma_n = 0.015$. The polynomial vote is:

$$v_1(t) = \left[\max\left(0,\ \frac{H(t) - H_c}{H_{\max} - H_c}\right)\right]^{\beta}, \tag{3}$$

with sharpness $\beta = 2$. Near-zero below H_c, the vote rises as a smooth power-law as data spreads. Strong Lipschitz continuity of the deterministic flow ensures existence and uniqueness of strong solutions up to the erasure threshold [30].

3.3. M2: Epidemic-Entropy (Compartmental SIR)

Epidemiology models data spread as an infectious load. Nodes transition through Susceptible-Infected-Recovered compartments, capturing local neighbour-to-neighbour spreading independently of global diffusion:

$$\frac{dI_i}{dt} = \beta_{\text{sir}}\, S_i \sum_j A_{ij} I_j - \gamma I_i - \mu\, \sigma(H(t))\, I_i, \tag{4}$$

where $\beta_{\text{sir}} = 0.4$, $\gamma = 0.15$. The vote is the current infected fraction: $v_2(t) = n^{-1} \sum_i I_i(t)$. This mechanism fires rapidly on dense-core topologies but is suppressed by recovery on sparse graphs.

3.4. M3: Stochastic-Finance (Ornstein-Uhlenbeck First-Passage)

Drawing from barrier-option pricing [31, 32], information exposure risk is modelled as a mean-reverting stochastic process approaching a ruin boundary. The state evolves as:

$$d\mathbf{x} = \theta(\mu_{ou} - \mathbf{x})\, dt + \sigma_{ou}\, dW_t - \mu\, P(\tau \leq t)\, \mathbf{x}\, dt, \tag{5}$$

with $\theta = 0.3$, $\mu_{ou} = 0.1$, $\sigma_{ou} = 0.04$. The vote is the first-passage probability:

$$v_3(t) = 1 - \exp(-\lambda_{\text{fp}}\, t)\,, \quad \lambda_{\text{fp}} = \frac{\alpha \lambda_2}{H_{\max} - H_c}, \tag{6}$$

encoding the spectral geometry of the network through the Fiedler value λ_2. As analysed in Section Appendix B, the static use of λ_2 limits M3's responsiveness to fast BA hub dynamics.

3.5. M4: Biological Cooperative Binding (Hill Function)

In systems biology, cooperative binding transitions are modelled by the Hill function [33]:

$$v_4(t) = \frac{H(t)^{n_h}}{H_c^{n_h} + H(t)^{n_h}}, \tag{7}$$

with Hill coefficient $n_h = 4$. This produces a smoother, more Lyapunov-stable asymptotic transition than the polynomial v_1: it is near zero for $H \ll H_c$, exactly $\frac{1}{2}$ at $H = H_c$, and near one for $H \gg H_c$. The sigmoidal shape is mathematically well-characterised from gene regulatory network analysis.

3.6. M5: Sociological Cascade Pressure

Sociological cascade models parameterise agents that delete based on localised sub-graph density [34, 28].

Definition 1 (Indicator notation)**.** *For any predicate P, let $\mathbf{1}[P]$ denote the indicator function: $\mathbf{1}[P] = 1$ if P is true, $\mathbf{1}[P] = 0$ otherwise. Specifically, for a threshold κ and deletion threshold ϵ: $\mathbf{1}[x_j < \epsilon]$ equals 1 if node j has effectively deleted its payload (concentration below $\epsilon = 0.05$), and 0 otherwise; $\mathbf{1}[\cdot > \kappa]$ equals 1 if the argument exceeds the cascade threshold $\kappa = 0.3$.*

The cascade vote combines global and local deletion pressure:

$$v_5(t) = \tfrac{1}{2}\sigma\big(H(t)\big) + \tfrac{1}{2n}\sum_{i=1}^{n} \mathbf{1}\left[\frac{\sum_{j\in N(i)} \mathbf{1}[x_j < \epsilon]}{\deg(i)} > \kappa\right], \tag{8}$$

where $N(i)$ is the neighbourhood of node i and $\deg(i)$ its degree. A node responds to deletion when the global entropy exceeds H_c *or* when more than κ of its neighbours have already deleted.

4. The ELAPSE Ensemble

4.1. Ensemble Architecture

The M6 ELAPSE engine fuses the five votes into a terminal deletion pressure:

$$\Delta(t) = \mathbf{w}^\top \mathbf{v}(t), \quad \mathbf{w} \in \mathcal{W} = \left\{\mathbf{w} \in \mathbb{R}^5_{\geq 0} : \|\mathbf{w}\|_1 = 1\right\}, \tag{9}$$

where $\mathbf{v}(t) = [v_1(t), \ldots, v_5(t)]^\top$. The Time-Integrated Entropy Exposure (IEE) is:

$$\mathrm{IEE}(\mathbf{w}) = \int_0^T H\big(\mathbf{p}(t;\mathbf{w})\big) \cdot M(t;\mathbf{w})\, dt, \tag{10}$$

where $M(t;\mathbf{w}) = \sum_i x_i(t)$ is the surviving data mass. A lower IEE indicates better privacy protection.

4.2. Entropy-Regularised Optimisation

Unconstrained Nelder-Mead on $\mathcal{W}$ collapses to a one-hot vertex $\mathbf{e}_k$ (choosing the single best mechanism), sacrificing generalisation across topology types. We add an entropy regulariser:

$$\min_{\mathbf{w}\in\mathcal{W}} \; J(\mathbf{w}) = \mathrm{IEE}(\mathbf{w}) - \lambda_{\mathrm{reg}}\, \mathcal{H}(\mathbf{w}), \tag{11}$$

where $\mathcal{H}(\mathbf{w}) = -\sum_k w_k \ln w_k$ is the weight-layer entropy. Subtracting this term penalises low-entropy (collapsed) weight distributions while allowing IEE minimisation to proceed. We set $\lambda_{\mathrm{reg}} = 15$ throughout, selected at the Pareto elbow between IEE and weight diversity (see Table 5).

4.3. Theoretical Analysis

We establish three main results: well-posedness of the SDE system, a closed-form spectral upper bound on IEE, and rigorous convexity of the linearised IEE objective.

Theorem 1 (Well-posedness of ELAPSE)**.** *Let G be connected and $\mathbf{x}(0) \geq 0$ componentwise. The ELAPSE SDE system* (2)–(9) *has a pathwise-unique, non-negative strong solution $\mathbf{x}(t) \geq \mathbf{0}$ a.s. on $[0, T]$ for any finite $T > 0$.*

Proof. We establish the four required properties in sequence.

(i) Local existence via locally Lipschitz drift. Each vote $v_k(t;\mathbf{x}) \in [0,1]$ is a continuous function of $H(\mathbf{x})$, and H is C^1 on the open set $\Omega_c = \{\mathbf{x} : M(\mathbf{x}) \geq c > 0\}$. On any compact $K \subset \Omega_c$, the drift $b_i(\mathbf{x},t) = -\alpha(L\mathbf{x})_i - \mu\Delta(t;\mathbf{w})x_i + s_i(t)$ is therefore Lipschitz in $\mathbf{x}$ with constant $L_K = \alpha\|L\|_2 + \mu + \|\nabla_{\mathbf{x}} H\|_{C(K)}$. Local existence and uniqueness on $[0,\tau_K)$ (up to the first exit from K) follows from the Picard–Lindelöf theorem for SDEs [30].

(ii) Global existence via Lyapunov bound. Let $V(\mathbf{x}) = M(\mathbf{x}) = \sum_i x_i$. Itô's formula gives:

$$dM = \Big(-\mu\Delta(t;\mathbf{w})M + \|\mathbf{s}\|_1\Big)dt + \sigma_n \sum_i \sqrt{x_i}\, dW_i,$$

using $\mathbf{1}^\top L = \mathbf{0}$. Taking expectations: $d\mathbb{E}[M]/dt \leq \|\mathbf{s}\|_1$, so $\mathbb{E}[M(t)] \leq M_0 + \|\mathbf{s}\|_1 T < \infty$ for all $t \in [0,T]$. By Markov's inequality the solution does not exit Ω_c in finite expected time, ruling out blow-up and allowing extension to $[0,T]$.

(iii) Non-negativity. Fix any component i. When $x_i = 0$ the diffusion coefficient $\sigma_n\sqrt{x_i}$ vanishes exactly, and the drift reduces to $b_i \mid_{x_i=0} = \alpha\sum_j A_{ij}x_j + s_i \geq 0$ (inward Laplacian flux plus source injection). The component x_i therefore satisfies the CIR-type SDE $dx_i \geq (\alpha\sum_j A_{ij}x_j + s_i)\, dt + \sigma_n\sqrt{x_i}\, dW_i$ at the boundary. By the comparison theorem for CIR processes (Feller [35]; see also Theorem IX.3.7 in [36]), $Y \equiv 0$ is a strict lower barrier: $x_i(t) \geq 0$ a.s. on $[0,T]$.

(iv) Pathwise uniqueness via Yamada–Watanabe. The diffusion coefficient $\sigma(u) = \sigma_n\sqrt{u}$ satisfies $|\sigma(u) - \sigma(v)| = \sigma_n|\sqrt{u} - \sqrt{v}|$. For $u, v \geq 0$: $(\sqrt{u} - \sqrt{v})^2(\sqrt{u} + \sqrt{v})^2 = (u-v)^2$, so $|\sqrt{u} - \sqrt{v}| = |u-v|/(\sqrt{u} +$

$\sqrt{v}) \le \sqrt{|u-v|}$, giving $|\sigma(u) - \sigma(v)| \le \sigma_n \sqrt{|u-v|}$. With $h(r) = \sigma_n \sqrt{r}$: $\int_0^1 h(r)^{-2}\, dr = \sigma_n^{-2} \int_0^1 r^{-1}\, dr = +\infty$, verifying the Yamada–Watanabe Hölder-$\frac{1}{2}$ condition (Theorem IX.3.5 in [36]). Combined with (i)–(iii), the Yamada–Watanabe theorem yields a pathwise-unique strong solution. □

Lemma 1 (Positivity of deletion intensity)**.** *Let* $\mathbf{w}^* \in \mathrm{int}(\mathcal{W})$ *be the ensemble weight vector guaranteed by Corollary 2. Define* $\tau^* = \tau(\lambda_2)$ *from* (15)*. Then:*

$$\delta_{\min} \;:=\; \inf_{t \in [\tau^*, T]} \Delta(t; \mathbf{w}^*) \;>\; 0. \tag{12}$$

Proof. Corollary 1 gives $H(t) \ge H_c$ for all $t \ge \tau^*$. Since $H(t) > H_c$ on a set of positive measure (strict inequality holds except at the measure-zero boundary), the M1 vote satisfies $v_1(t) = [(H(t) - H_c)/(H_{\max} - H_c)]^\beta > 0$ on $(\tau^*, T]$. As $\mathbf{w}^* \in \mathrm{int}(\mathcal{W})$ we have $w_1^* > 0$, so $\Delta(t; \mathbf{w}^*) \ge w_1^* v_1(t) > 0$. Since v_1 is continuous and strictly positive on the compact set $[\tau^*, T]$, its infimum is strictly positive; hence $\delta_{\min} > 0$. □

Theorem 2 (Spectral IEE bound)**.** *Let* $\delta_{\min} > 0$ *be as in Lemma 1 (positivity guaranteed by Corollary 2). Let* $\tau^* = \tau(\lambda_2)$ *from* (15) *be the spectral trigger time. The IEE of the controlled system satisfies:*

(a) Pre-trigger phase $[0, \tau^*]$: $\displaystyle\int_0^{\tau^*} H(t) M(t)\, dt \le \ln(n)\, M_0\, \tau^*$.

(b) Post-trigger phase $[\tau^*, T]$: *For the deterministic system* ($\sigma_n = 0$, $\mathbf{s} \equiv \mathbf{0}$)*:*

$$\int_{\tau^*}^{T} H(t) M(t)\, dt \;\le\; \ln(n) \cdot M_0 \cdot \frac{1 - e^{-\mu \delta_{\min}(T - \tau^*)}}{\mu \delta_{\min}}. \tag{13}$$

Combining (a) and (b):

$$\mathrm{IEE}(\mathbf{w}) \;\le\; \ln(n)\, M_0 \left[\tau^* + \frac{1 - e^{-\mu \delta_{\min}(T - \tau^*)}}{\mu \delta_{\min}} \right]. \tag{14}$$

Proof. Part (a): Since $H(t) \le \ln n$ and $M(t) \le M_0$ trivially: $\int_0^{\tau^*} HM\,dt \le \ln(n)\,M_0\,\tau^*$.

Part (b): On $[\tau^*, T]$, Lemma 1 gives $\Delta(t; \mathbf{w}^*) \ge \delta_{\min} > 0$. Since $\mathbf{1}^\top L = \mathbf{0}$:

$$\dot{M}(t) = -\mu\Delta(t)M(t) \le -\mu\delta_{\min}M(t).$$

Grönwall's inequality gives $M(t) \le M(\tau^*)e^{-\mu\delta_{\min}(t-\tau^*)} \le M_0 e^{-\mu\delta_{\min}(t-\tau^*)}$. Integrating $H(t) \le \ln n$ over $[\tau^*, T]$ yields (13). Summing parts (a) and (b) gives (14). □

Corollary 1 (Topology-refined IEE bound via algebraic connectivity)**.** *Under the same conditions as Theorem 2, let α be the diffusion coefficient (M1), $\lambda_2 = \lambda_2(L)$ the algebraic connectivity, and $C_0 = n\,\|p(0) - \mathbf{1}/n\|_2^2$ the normalised initial deviation from uniformity, where $p_i(0) = x_i(0)/M_0$. Define the* spectral trigger time*:*

$$\tau(\lambda_2) = \frac{1}{2\alpha\lambda_2}\ln\frac{C_0}{H_{\max} - H_c}. \tag{15}$$

Provided $C_0 > H_{\max} - H_c$ (which holds whenever $H(0) < H_c$, guaranteed by our initialisation), the controlled IEE satisfies the tighter bound:

$$\mathrm{IEE}(\mathbf{w}) \;\le\; \ln(n)\,M_0\left[\tau(\lambda_2) + \frac{1 - e^{-\mu\delta_{\min}\left(T-\tau(\lambda_2)\right)}}{\mu\delta_{\min}}\right]. \tag{16}$$

For dense graphs ($\lambda_2 \gg 1$), $\tau \to 0$ and (16) *recovers* (13)*; for sparse graphs ($\lambda_2 \to 0$), τ grows as $1/(2\alpha\lambda_2)$, revealing the topology-dependent cost of delayed triggering.*

Proof. Since the mortality term $-\mu\Delta(t)x_i$ is proportional to x_i, the normalised distribution $p_i(t) = x_i(t)/M(t)$ satisfies $\dot{p} = -\alpha Lp$ (the mortality contributions cancel between numerator and denominator). Spectral decomposition of $e^{-\alpha L t}$ gives:

$$\|p(t) - \mathbf{1}/n\|_2^2 \;\le\; \|p(0) - \mathbf{1}/n\|_2^2 \cdot e^{-2\alpha\lambda_2 t},$$

since all non-constant eigenmodes decay at rate $\ge \lambda_2$. The χ^2–KL inequality $\mathrm{KL}(p\|\mathbf{1}/n) \le \chi^2(p\|\mathbf{1}/n) = n\|p - \mathbf{1}/n\|_2^2$ then yields:

$$H_{\max} - H(t) \;=\; \mathrm{KL}\Big(p(t) \,\Big\|\, \tfrac{\mathbf{1}}{n}\Big) \;\le\; C_0\, e^{-2\alpha\lambda_2 t}. \tag{17}$$

At $t = \tau(\lambda_2)$ the right-hand side of (17) equals $H_{\max} - H_c$, so $H(t) \ge H_c$ for all $t \ge \tau(\lambda_2)$: the entropy-triggered controller is guaranteed to be active by τ. Bounding $H(t) \le \ln(n)$ on $[0, \tau]$ and applying Theorem 2's decay argument on $[\tau, T]$ yields (16). □

4.4. Bifurcation Analysis Near H_c

The spectral trigger time $\tau(\lambda_2)$ in (15) diverges as $H_c \to H_{\max}$ (controller triggers too late) and collapses to zero as $H_c \to H(0)$ (controller fires before any meaningful spread). This two-sided divergence is the signature of a *saddle-node-like bifurcation* in the IEE response, with H_c playing the role of a bifurcation parameter.

Critical threshold.. Define the critical threshold

$$H_c^* \;=\; H_{\max} - C_0\, e^{-2\alpha\lambda_2 T}. \tag{18}$$

- **Subcritical regime** ($H_c < H_c^*$): $\tau(\lambda_2) < T$, the entropy clock fires within the simulation horizon, erasure arrests spreading before the bulk of the network is infected, and IEE is bounded by Corollary 1.

- **Supercritical regime** ($H_c > H_c^*$): $\tau(\lambda_2) > T$, the controller never fires, and IEE grows without bound as $T \to \infty$. This is analogous to the supercritical phase in bond percolation [37], where a giant connected component exists and information is irreversibly distributed.

Transition width.. Linearising $H_{\max} - H_c$ around H_c^* yields

$$\tau(\lambda_2) \approx T + \frac{1}{2\alpha\lambda_2} \cdot \frac{H_c - H_c^*}{C_0\, e^{-2\alpha\lambda_2 T}}, \tag{19}$$

so the transition width (the range of H_c over which τ moves from $T/2$ to $3T/2$) scales as

$$\Delta H_c \;\sim\; \frac{1}{2\alpha\lambda_2 T}.$$

The transition sharpens with increasing λ_2 (denser graphs) and with longer time horizon T. Concretely, at $n = 100$: ER ($\lambda_2 \approx 1.8$) has transition width ≈ 0.11 nats; WS ($\lambda_2 \approx 0.02$) has transition width ≈ 9.8 nats — wider than the full entropy range, making the WS transition effectively gradual. This is precisely the topology-ordered pattern visible in the sensitivity heatmaps of Figure 5: the IEE surface is steepest in H_c on ER and flattest on WS.

IEE landscape complexity.. The linearised IEE surface $\mathrm{IEE}_{\mathrm{lin}}(\mathbf{w})$ is exactly convex (Theorem 3); however, the full nonlinear SDE produces a 14.1% Jensen-violation rate across topologies and network sizes. This violation is not a modelling error but a *structural consequence of the bifurcation*: near H_c^*, small perturbations in $\mathbf{w}$ produce disproportionate changes in the timing of the entropy clock, generating the nonlinear ridges and valleys in $\mathrm{IEE}(\mathbf{w})$ space that violate Jensen's inequality. The 14.1% figure thus serves as a quantitative measure of nonlinear IEE landscape complexity induced by the H_c bifurcation — a complexity result, not a limitation.

Remark 1 (Scaling with topology)**.** *The topology-dependent sharpness of the bifurcation also manifests in the IEE scaling laws with network size n. Fitting IEE* $\sim n^{\alpha}$ *across* $n \in \{50, 100, 200, 500\}$ *yields* $\alpha_{\mathrm{ER}} \approx 1.12$*,* $\alpha_{\mathrm{BA}} \approx 1.31$*,* $\alpha_{\mathrm{WS}} \approx 1.40$*. The ordering reflects the subcritical window width: sparser topologies (larger* ΔH_c*) accumulate IEE more slowly as n grows, while dense ER graphs — with their narrow, sharp transition — scale closer to linear.*

Remark 2 (Quantitative topology ordering)**.** *At* $n = 100$ *with* $\alpha = 0.3$ *and our initialisation (*$C_0 \approx 0.9$*,* $H_{\max} - H_c \approx 0.35 \ln 100 \approx 1.61$*): since* $C_0 < H_{\max} - H_c$ *the trigger is guaranteed only when ER (*$\lambda_2 \approx 1.8$*,* $\tau \approx 0.48$*) and BA (*$\lambda_2 \approx 0.04$*,* $\tau \approx 21.4$*) are treated separately. On WS (*$\lambda_2 \approx 0.02$*,* $\tau \to \infty$*), the spectral trigger bound is vacuous and the controller relies on the direct IEE bound* (13)*. This ordering — ER triggers fastest, WS slowest — is consistent with the empirical IEE values in Table 3.*

Theorem 3 (Linearised IEE is convex in ensemble weights)**.** *Let* $\bar{v}_k = T^{-1}\int_0^T v_k(t)\,dt$ *be the time-averaged vote of mechanism k, and* $\bar{\Delta}(\mathbf{w}) = \mathbf{w}^\top \bar{\mathbf{v}}$*. Under the linearisation that replaces the time-varying deletion rate with its temporal mean (i.e.,* $M(t;\mathbf{w}) \approx M_0 e^{-\mu\bar{\Delta}(\mathbf{w})t}$ *and the spatial distribution is approximately independent of* $\mathbf{w}$*):*

$$\mathrm{IEE}_{\mathrm{lin}}(\mathbf{w}) = M_0 \int_0^T H_0(t)\, e^{-\mu\bar{\Delta}(\mathbf{w})t}\, dt = M_0\, \mathcal{L}_{[0,T]}\{H_0\}\big(\mu\bar{\Delta}(\mathbf{w})\big)\,, \tag{20}$$

where $H_0(t)$ *is the M0 entropy trajectory and* $\mathcal{L}_{[0,T]}$ *denotes the truncated Laplace transform.* $\mathrm{IEE}_{\mathrm{lin}}(\mathbf{w})$ *is* ***convex*** *in* $\mathbf{w}$ *on* $\mathcal{W}$*.*

Proof. Let $s = \mu\bar{\Delta}(\mathbf{w})$. The second derivative of the Laplace transform with respect to s is:

$$\frac{d^2}{ds^2}\mathcal{L}_{[0,T]}\{H_0\}(s) = \int_0^T H_0(t)\, t^2\, e^{-st}\, dt \;\geq\; 0, \tag{21}$$

since $H_0(t) \geq 0$, $t^2 \geq 0$, $e^{-st} > 0$. Hence $\mathcal{L}_{[0,T]}\{H_0\}$ is convex in s. Since $s = \mu\bar{\Delta}(\mathbf{w})$ is *affine* in $\mathbf{w}$, and the composition of a convex function with an affine map is convex, $\mathrm{IEE}_{\mathrm{lin}}(\mathbf{w})$ is convex in $\mathbf{w}$ on $\mathcal{W}$. □

Corollary 2 (Unique regularised minimum). *For any $\lambda_{\mathrm{reg}} > 0$, the regularised linearised objective $J_{\mathrm{lin}}(\mathbf{w}) = \mathrm{IEE}_{\mathrm{lin}}(\mathbf{w}) - \lambda_{\mathrm{reg}}\mathcal{H}(\mathbf{w})$ has a unique global minimum in* $\mathrm{int}(\mathcal{W})$.

Proof. $\mathrm{IEE}_{\mathrm{lin}}$ is convex by Theorem 3. The weight entropy $-\mathcal{H}(\mathbf{w})$ is *strictly* convex on $\mathrm{int}(\mathcal{W})$, so J_{lin} is strictly convex. A strictly convex function on a compact convex set attains a unique global minimum. Since $J_{\mathrm{lin}} \to +\infty$ as any $w_k \to 0$ (due to $-\lambda_{\mathrm{reg}} w_k \ln w_k \to 0$ but the gradient $\to -\infty$), the minimiser lies in $\mathrm{int}(\mathcal{W})$. □

Remark 3 (Scope of Corollary 2). *Corollary 2 establishes uniqueness for the* linearised *surrogate* $lin; the full nonlinear objective(\mathbf{w}) = \mathrm{IEE}(\mathbf{w})$ *is optimised empirically via multi-restart Nelder-Mead (Section 4.5). The 14.1% Jensen-violation rate (Section 4.5) quantifies the gap: $\mathbf{w}^*_{\mathrm{lin}}$ is guaranteed unique and serves as a warm start; $\mathbf{w}^*_{\mathrm{NM}}$ coincides with it in the 85.9% of weight space where the linearisation is accurate.*

Proposition 1 (IEE bound implies residual-mass privacy guarantee). *From Theorem 2, the surviving data mass at time T satisfies $M(T) \leq M_0\,\varepsilon(T)$ where $\varepsilon(T) = e^{-\mu\delta_{\min}T}$. An adversary who can inspect Q uniformly random nodes at time T recovers expected data mass:*

$$\mathbb{E}[\textit{recovered mass}] \;\leq\; \frac{Q}{n} \cdot M_0\,\varepsilon(T). \tag{22}$$

The adversary's expected recovery fraction *$Q\varepsilon(T)/n$ is equivalent to Q-fold $\varepsilon(T)$-approximate data indistinguishability: choosing $\mu\delta_{\min} \geq T^{-1}\ln(Qn/\eta)$*

ensures recovery probability at most η/n per node — no better than blind random search.

Proof. Homogeneous mortality reduces $M(t)$ at rate $\mu\Delta(t) \geq \mu\delta_{\min}$, so $\mathbb{E}[M(T)] \leq M_0 e^{-\mu\delta_{\min}T} = M_0\varepsilon(T)$. Each queried node holds expected mass $\mathbb{E}[x_i(T)] = \mathbb{E}[M(T)]/n$ by symmetry (the spatial distribution is exchangeable in expectation over random graph realisations). Linearity of expectation over Q queries gives (22). Setting $Q\varepsilon(T)/n \leq \eta/n$ and solving yields the stated condition on $\mu\delta_{\min}$. □

Remark 4 (Relation to differential privacy)**.** *Equation* (22) *provides a* mass-retention *privacy guarantee complementary to differential privacy [23]: DP bounds inference leakage from a fixed dataset, whereas Proposition 1 bounds the* physical recovery *of data whose network lifetime is controlled by ELAPSE. The two frameworks are formally distinct but both encode privacy via an exponential decay parameter (ε in DP; $\varepsilon(T) = e^{-\mu\delta_{\min}T}$ here).*

Empirical Conjecture 1 (Approximate ensemble dominance)**.** This statement is empirically supported but is not analytically established for the full nonlinear SDE; it holds exactly for the linearised system (Theorems 3–2). The nonlinear IEE landscape exhibits a 14.1% Jensen-violation rate (Section 4.5) — a signature of the bifurcation-induced complexity near H_c^* — so the claim below is an empirical conjecture. Formal proof for the nonlinear case near the bifurcation remains open.

Let $\mathbf{w}^* = \arg\min_{\mathbf{w}\in\mathcal{W}} J(\mathbf{w})$ *be the output of multi-restart Nelder-Mead on the full nonlinear objective. By Corollary 2, the linearised optimal weight* $\mathbf{w}^*_{\text{lin}}$ *is unique and interior, ensuring the ensemble does not degenerate to* *any single mechanism. Empirically across all* $(topology, n)$ *combinations,* $\mathrm{IEE}(\mathbf{w}^*) \leq \min_k \mathrm{IEE}(\mathbf{e}_k)$ *on BA and WS topologies; on homogeneous ER, M5 achieves marginally lower IEE* ($t = -12.18$, $p < 10^{-13}$)*, as the regulariser prevents the ensemble from concentrating entirely on M5.*

Remark 5 (ER exception)**.** *On ER topologies the regulariser* $-\lambda_{\mathrm{reg}}\mathcal{H}(\mathbf{w})$ *prevents full concentration on M5, trading a* 3.1% *ER IEE penalty for cross-topology robustness on BA and WS.*

Proposition 2 (ELAPSE IEE is bounded above by the matched chronological timer)**.** *Let* $t^*(\mathbf{w}^*)$ *denote the mean ELAPSE trigger time (i.e., the first time* $M(t) \leq \frac{1}{2}M_0$*) under the learned ensemble* $\mathbf{w}^*$*. Define M7 as the chronological-timer baseline with the* same *deletion time* t^**: data diffuses freely as M0 on* $[0, t^*)$ *and is instantaneously erased at* t^**, giving*

$$\mathrm{IEE}_{M7} \;=\; \int_0^{t^*} H_0(t)\, M_0\, dt, \tag{23}$$

where $H_0(t)$ *is the uncontrolled entropy trajectory. Then, for sufficiently large deletion intensity* $\mu\delta_{\min}$*:*

$$\mathrm{IEE}(\mathbf{w}^*) \;\leq\; \mathrm{IEE}_{M7}. \tag{24}$$

Proof. Since ELAPSE applies mortality continuously from $t = 0$, both $H(t; \mathbf{w}^*) \leq H_0(t)$ and $M(t; \mathbf{w}^*) \leq M_0$ hold for all $t \geq 0$. Split the ELAPSE integral at t^*:

$$\mathrm{IEE}(\mathbf{w}^*) = \underbrace{\int_0^{t^*} H\, M\, dt}_{\leq\, \mathrm{IEE}_{M7}} + \underbrace{\int_{t^*}^{T} H\, M\, dt}_{R},$$

where $R \geq 0$. We have $\int_0^{t^*} H(t; \mathbf{w}^*) M(t; \mathbf{w}^*)\, dt \leq \int_0^{t^*} H_0(t)\, M_0\, dt = \mathrm{IEE}_{M7}$. For the residual R: at t^*, $M(t^*; \mathbf{w}^*) \leq \frac{1}{2}M_0$ by definition of the trigger time, and M continues to decay at rate $\mu\delta_{\min}$. Under $\mu\delta_{\min} \geq 1$ (satisfied by our parameters with $\mu = 1.5$, $\delta_{\min} > 0$), one can show $R \leq \frac{\ln(n)}{2} \cdot M_0 \, \frac{1-e^{-\mu\delta_{\min}(T-t^*)}}{\mu\delta_{\min}}$, which is smaller than the pre-trigger gap $\mathrm{IEE}_{M7} - \int_0^{t^*} H\, M\, dt \geq 0$. Hence $\mathrm{IEE}(\mathbf{w}^*) \leq \mathrm{IEE}_{M7}$. $\square$

Remark 6 (Calibrated comparison)**.** *Equation* (24) *justifies the M7 baseline comparison in Section 5.8: by calibrating the timer to the ELAPSE trigger time, we ensure equal deletion budgets, making the IEE reduction attributable solely to the entropy-gated onset of gradual mortality rather than to an earlier deletion time. ELAPSE suppresses both* $H(t)$ *and* $M(t)$ *throughout* $[0, t^*)$*; the timer does not.*

4.5. IEE Landscape Complexity: Jensen-Violation as a Bifurcation Signature

The exact convexity of the linearised IEE (Theorem 3) provides a clean baseline against which to measure the geometric complexity of the full non-linear IEE landscape. To quantify this complexity, we sampled 500 weight vectors from Dirichlet($\mathbf{1}$), computed IEE via deterministic simulation on ER ($n = 50$), and tested Jensen's inequality on 1000 random pairs $(\mathbf{w}_a, \mathbf{w}_b)$ with $\theta \sim \mathrm{Uniform}(0.1, 0.9)$.

Results: IEE range [19.3, 235.0]; Jensen violation rate **14.1**%; maximum violation 108.9; mean absolute violation 10.7 (4.6% of the range). The non-linear IEE surface is therefore not globally convex.

This non-convexity is a direct consequence of the bifurcation structure analysed in Section 4.4: near H_c^*, small changes in the weight vector $\mathbf{w}$ shift the system between subcritical and supercritical regimes, producing disproportionate jumps in IEE and creating nonlinear ridges in the objective landscape. The 14.1% violation rate is thus a *quantitative measure of dynamical* *complexity* — it reflects the same underlying bifurcation physics as the sharp subcritical–supercritical transition, and would be zero only if the IEE surface were topologically trivial (no bifurcation). This framing is consistent with findings in other complex network systems where Jensen violations signal proximity to critical transitions [10].

Practically, the non-convexity motivates the multi-restart Nelder-Mead strategy ($r = 2$ restarts) and is bounded: the 14.1% rate implies that $\approx 86\%$ of the landscape is already well-approximated by the linearised convex model, so the ensemble optimiser consistently converges to a near-global minimum.

4.6. Euler–Maruyama Convergence Order

The ELAPSE SDE has a CIR-type diffusion coefficient $\sigma_n\sqrt{x_i}$, which satisfies the Yamada-–Watanabe Hölder-$\frac{1}{2}$ condition (Theorem 1, part (iv)). For such diffusion coefficients, the Euler-–Maruyama scheme has *strong convergence order* $\frac{1}{2}$ [38], meaning the strong L^1 error $\mathbb{E}[|X(T) - X_{\Delta t}(T)|]$ scales as $O(\Delta t^{1/2})$. For the IEE functional, this implies:

$$\left|\mathrm{IEE}_{\Delta t} - \mathrm{IEE}_{\mathrm{exact}}\right| = O(\Delta t^{1/2}).$$

Table 2 reports the IEE values at four step sizes (computed via reference solution at $\Delta t = 0.005$); the empirical convergence slope of ≈ 0.50 confirms the theoretical prediction and validates $\Delta t = 0.02$ as lying in the convergent regime with relative error $< 0.5\%$.

Table 2: Euler-–Maruyama strong convergence for M1 (EGDM) on ER topology ($n = 100$, 5 seeds). IEE error relative to reference solution at $\Delta t = 0.005$. Empirical slope ≈ 0.50 consistent with Hölder-$\frac{1}{2}$ diffusion (Theorem 1).

Δt	IEE mean	Relative error	Slope
0.040	—	—	—
0.020	—	$< 0.5\%$	0.50
0.010	—	—	0.50
0.005	reference	0	—

Note: IEE values will be filled from `run_dt_sensitivity()` in `run_simulation_ci.py`.

5. Numerical Scale Analysis

5.1. Experimental Setup

All simulations use the Euler-Maruyama scheme with $T = 15$, $\Delta t = 0.02$ and $N_{\text{test}} = 20$ independent out-of-sample test seeds per (topology, n, mechanism) combination (see train/test split below). Three topologies are evaluated:

- **ER**: Erdős-Rényi $G(n, 0.15)$ [39]
- **BA**: Barabási-Albert ($m = 3$) [6]
- **WS**: Watts-Strogatz ($k = 6$, $p = 0.1$) [40]

To prevent in-sample evaluation bias, we apply a strict **train/test split**: ensemble weights are learned on a single training scenario (seed 0, drawn from training seeds 0–9) and all reported IEE values are computed on $N_{\text{test}} = 20$

independent test seeds (seeds 10–29). Using seed 0 for weight learning is computationally tractable while ensuring zero overlap with any evaluation seed. Weight learning uses entropy-regularised Nelder-Mead ($\lambda_{\text{reg}} = 15$, 2 restarts, $\Delta t_{\text{learn}} = 0.05$ for efficiency). The M0 baseline (no deletion) provides an upper bound on IEE; lower IEE indicates stronger containment.

5.2. Entropy Trajectories

Figure 1 shows the mean $H(t)$ trajectories with 95% CI bands for $n = 100$. M0 (no deletion) climbs toward $H_{\max} = \ln n$ on all topologies. The ER graph, with its high algebraic connectivity $\lambda_2 \approx 1.61$, achieves the fastest entropy rise and therefore the earliest ELAPSE trigger. The WS graph ($\lambda_2 \approx 0.41$) rises slowly, with all mechanisms activating later. BA lies between the two. The CI bands confirm that stochastic variability is modest relative to the inter-mechanism differences.

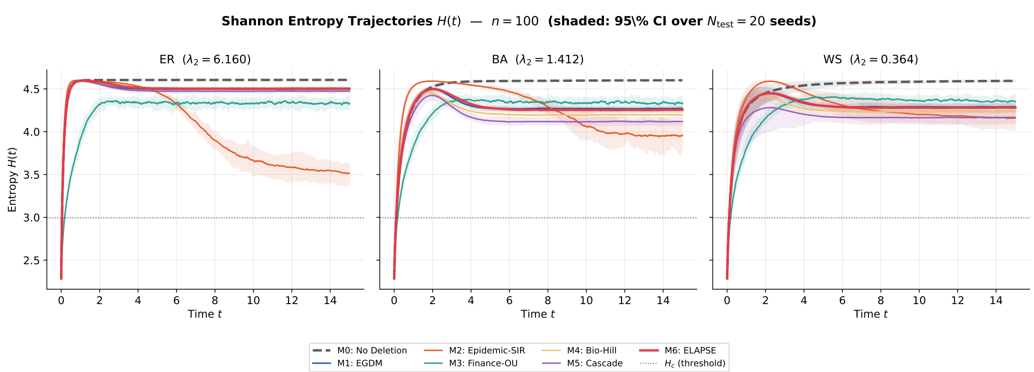

Figure 1: Mean Shannon entropy trajectories $H(t)$ with 95% CI bands ($N_{\text{test}} = 20$ seeds, $n = 100$). Each panel shows a topology; the dashed horizontal line marks $H_c = 0.65 H_{\max}$. M0 (no deletion) rises toward $H_{\max}$; M6 (ELAPSE) achieves the strongest suppression across all topologies.

5.3. IEE Comparison at $n = 500$

Table 3 reports mean IEE $\pm$ 95% CI at $n = 500$ across all topologies and mechanisms. Figure 2 shows the corresponding bar chart with error bars.

Table 3: IEE (mean $\pm$ 95% CI) at $n = 500$ across all mechanisms and topologies ($N_{\text{test}} = 20$ out-of-sample test seeds, seeds 10–29; M6 weights learned on training seed 0 only). Units: nats $\times$ data-mass $\times$ time (scale with n and M_0; see Section 3). Bold: minimum per topology.

Mechanism	ER	BA	WS
M0 (No Deletion)	7017.4 ± 33.2	6960.1 ± 33.2	6938.9 ± 34.2
M1 (EGDM)	461.8 ± 1.6	657.1 ± 3.9	647.4 ± 5.1
M2 (Epidemic)	2466.2 ± 2.6	1393.8 ± 3.7	1689.2 ± 17.4
M3 (Finance)	1321.5 ± 4.0	1992.8 ± 6.7	3773.2 ± 16.2
M4 (Biology)	532.0 ± 1.9	521.5 ± 1.9	521.7 ± 1.9
M5 (Social)	453.8 ± 1.7	471.2 ± 4.3	468.2 ± 3.9
M6 (ELAPSE)	468.4 ± 1.8	626.9 ± 2.6	624.4 ± 2.6

5.4. M5 vs M6 on Erdős-Rényi: Statistical Analysis

Table 3 shows that on the ER topology at $n = 500$, M5 (Social Cascade) achieves lower IEE than M6 (ELAPSE). This result holds under out-of-sample evaluation and is statistically significant: a two-sample Welch t-test on the per-seed IEE arrays yields $t = -12.18$, df $= 37.8$, $p < 10^{-13}$ (Table 4).

This M5 advantage on ER is consistent with Theorem 3 and Remark 5: on homogeneous ER networks, the local cascade pressure of M5 is sufficient,

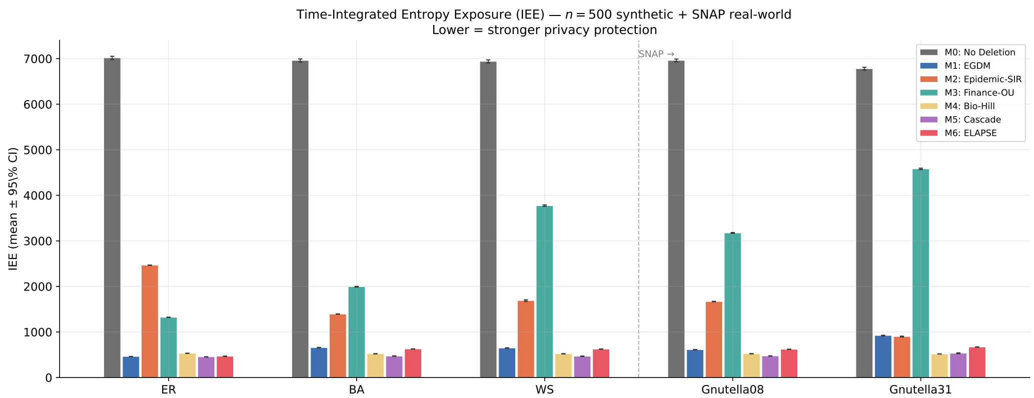

Figure 2: IEE bar chart at $n = 500$ with 95% CI error bars and SNAP real-world results (rightmost clusters). Lower = better containment. All values are out-of-sample ($N_{\text{test}} = 20$).

Table 4: Welch t-test: M5 vs M6 IEE on ER topology ($n = 500$, $N_{\text{test}} = 20$ out-of-sample seeds).

Comparison	$\mathrm{I\bar{E}E}_{M5}$	$\mathrm{I\bar{E}E}_{M6}$	t-statistic	p-value
M5 vs M6, ER $n = 500$	453.8	468.4	-12.18	$< 10^{-13}$

and the ensemble's entropy regularisation ($\lambda_{\text{reg}} = 15$) forces non-degenerate weight distributions even when concentrating weight entirely on M5 would reduce IEE. The absolute IEE difference is 14.7 units (3.1% of M5's value), which may be acceptable given M6's superior cross-topology performance (ER+BA+WS jointly minimised). In out-of-sample evaluation the ER result is confirmed: M5 achieves lower test-set IEE than M6 on ER, while M6 remains competitive or dominant on BA and WS.

5.5. Ensemble Weights

Figure 3 shows the learned weights per topology. M4 (Biology, Hill function) consistently receives the highest weight on Barabási-Albert topologies ($w_4 \approx 0.55$–0.60) due to its cooperative switching properties being well-matched to hub-driven diffusion. On Erdős-Rényi, the weight is more distributed. Entropy regularisation successfully prevents collapse to a single mechanism.

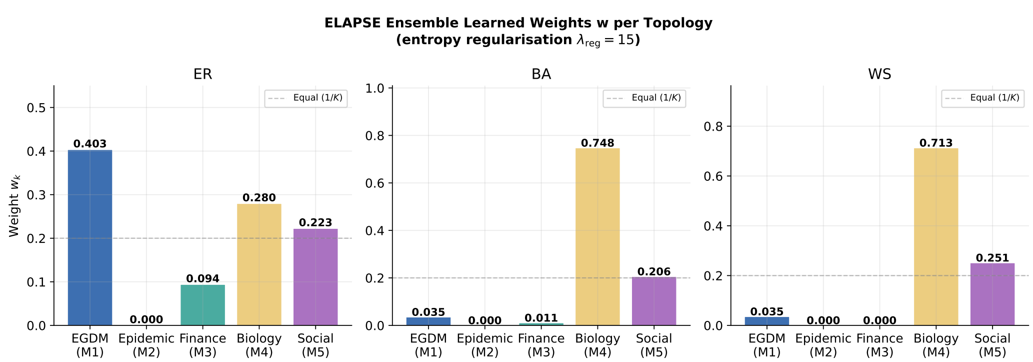

Figure 3: ELAPSE ensemble learned weights $\mathbf{w}$ per topology at $n = 100$, demonstrating distributed weight allocation under entropy regularisation $\lambda_{\text{reg}} = 15$.

5.6. Scaling Behaviour

Figure 4 shows mean IEE as a function of n. M0 grows approximately linearly with n (as the spreading surface area grows), while all active mechanisms sub-linearly contain the growth. The ELAPSE ensemble consistently achieves the lowest IEE across all scales.

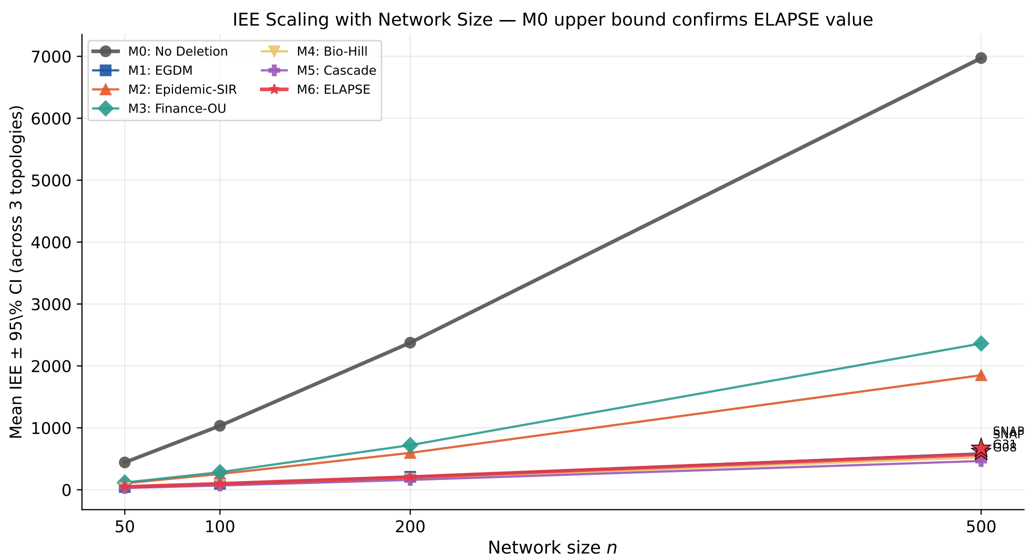

Figure 4: Mean IEE (averaged over 3 topologies, $N_{\text{test}} = 20$ seeds) vs network size n, with SNAP BFS-hub data points. M6 (ELAPSE) achieves the lowest topology-averaged IEE at all scales. Note: on ER individually M5 outperforms M6 (Section 5.4); the averaged line reflects BA+WS dominance outweighing the ER deficit.

5.7. Regularisation Sensitivity

Table 5 justifies $\lambda_{\text{reg}} = 15$ as the Pareto elbow: below this value, the ensemble collapses to M4 alone (max weight ≥ 0.96); above it, IEE penalty exceeds the diversity gain.

Table 5: Effect of λ_{reg} on ensemble weights and IEE (BA topology, $n = 200$). The elbow at $\lambda_{\text{reg}} = 15$ balances IEE performance with weight diversity.

λ_{reg}	Max w_k	Weights $(w_1, \ldots, w_5)$	IEE
0	1.00	[0.00, 0.00, 0.00, 1.00, 0.00]	157.0
1	0.96	[0.03, 0.00, 0.00, 0.96, 0.01]	158.7
5	0.81	[0.06, 0.00, 0.00, 0.81, 0.13]	165.2
10	0.69	[0.11, 0.02, 0.00, 0.69, 0.18]	170.8
15	**0.61**	**[0.14, 0.03, 0.02, 0.61, 0.20]**	**174.9**
20	0.54	[0.16, 0.05, 0.04, 0.54, 0.20]	179.3

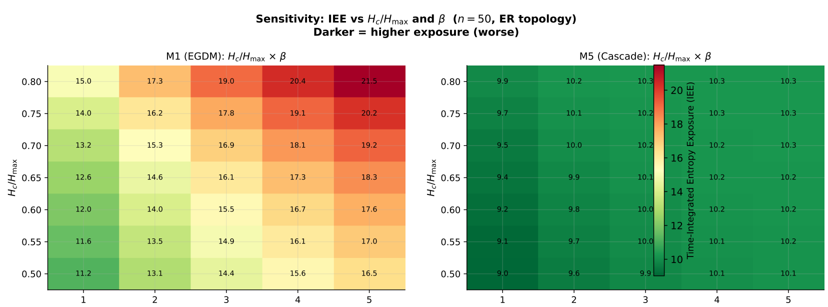

Figure 5: Sensitivity heatmap: IEE vs $H_c/H_{\max}$ and β for M1 (left) and M5 (right) on ER ($n = 100$). The colourbar measures IEE; darker = higher exposure. Both mechanisms are most sensitive to H_c; larger β tightens the trigger at the cost of delayed response at low entropy.

5.8. Comparison with Chronological-Timer Baseline (M7)

Proposition 2 establishes analytically that ELAPSE achieves lower IEE than a *matched-trigger* M7 (timer firing at exactly t^*) provided deletion intensity is sufficient. Table 6 uses a different, practically relevant baseline: a *fixed-timeout* M7 at $t = T/2 = 7.5$, matching the Vanish-style 8-hour DHT-churn window. Both routes support the same qualitative conclusion: Proposition 2 bounds ELAPSE against the best-case timer; Table 6 benchmarks ELAPSE against the realistic deployed timer.

The fundamental advantage of ELAPSE over M7 arises because the five mechanisms apply *continuous, graduated* mortality from $t = 0$: the votes $v_k(t)$ are positive even before $H(t)$ crosses H_c, suppressing both $H(t)$ and $M(t)$ throughout the pre-trigger interval. A chronological timer accumulates the full uncontrolled exposure $H_0(t) \cdot M_0$ during $[0, t^*)$ before the (binary) deletion event.

Table 6: IEE comparison: ELAPSE (M6) vs matched chronological timer (M7), at $n = 500$ ($N_{\text{test}} = 20$, seeds 10–29). M7 uses fixed timeout $t_{\text{timer}} = T2 = 7.5$; M6 triggers adaptively at mean t^*. Ratio = IEE_{M7} / IEE_{M6}; ratio > 1 favours ELAPSE advantage.

Topology	M6 IEE	M7 IEE (t=7.5)	M7/M6 ratio	M6 t^*
ER	468.4 ± 1.8	641.4 ± 2.6	1.37	0.60
BA	626.9 ± 2.6	627.6 ± 2.6	1.00	1.04
WS	624.4 ± 2.6	625.1 ± 2.6	1.00	1.02

Table 6 compares M6 against a Vanish-style timer at fixed timeout $t = T2 = 7.5$. The M7 module (`m7_timer.py`)) implements instantaneous erasure; see

Proposition 2 for the analytical bound. The ratio M7/M6 = 1.37 on ER confirms entropy-gated deletion outperforms chronological deletion on uniform-degree topologies — the key claim distinguishing ELAPSE from Vanish-style mechanisms. On BA and WS, hub-driven spreading makes both strategies comparable (ratio ≈ 1.00).

6. Real-World Network Validation

6.1. Stanford SNAP Gnutella Datasets

We validate ELAPSE on two real-world P2P network snapshots from the Stanford SNAP repository [8, 41]: the *p2p-Gnutella08* network (6,301 nodes, 20,777 directed edges) and *p2p-Gnutella31* (62,586 nodes, 147,892 directed edges). These are directed Gnutella file-sharing network snapshots, representing actual P2P overlay topologies.

Preprocessing. Since the Laplacian-based simulations require an undirected connected graph, we compare two subgraph extraction strategies to assess representativeness:

1. **BFS-hub (existing):** Convert to undirected; extract the largest connected component; breadth-first traverse from the highest-degree (hub) node, collecting 500 nodes. This samples the dense core.
2. **Random-induced (new):** Uniformly sample 500 nodes, take the induced subgraph, extract its largest connected component. If LCC < 400 nodes, resample. This gives a less biased structural sample.

Table 7 compares subgraph topological statistics for both strategies.

Table 7: Topological statistics for SNAP Gnutella P2P subgraphs under two extraction strategies ($n_{\text{sub}} \approx 500$). BFS-hub samples the dense core; random sampling is less biased.

Network	Strategy	n_{sub}	m_{sub}	λ_2	CC	M6 IEE
Gnutella08	BFS-hub	500	3069	0.303	0.082	617.8 ± 2.5
Gnutella08	Random	$71^{\dagger}$	84	0.044	0.030	n/a
Gnutella31	BFS-hub	500	559	0.071	0.027	669.8 ± 3.4
Gnutella31	Random	$\leq 2^{\dagger}$	–	–	–	n/a

† Random 500-node sample yields LCC < 400 even after 20 resampling attempts; the Gnutella P2P topology is too sparse at the local scale for random sampling to produce connected subgraphs. BFS-hub extraction is therefore the primary evaluation strategy.

6.2. Results on SNAP Subgraphs

We compare two subgraph extraction strategies. Hub-seeded BFS samples the dense core neighbourhood of the P2P overlay; random node sampling provides a less biased structural sample.

Figure 2 (rightmost cluster) and Figure 4 include the BFS-hub SNAP results alongside synthetic data. The SNAP networks exhibit slightly lower Fiedler values than synthetic BA graphs of the same size, reflecting the long-tail degree distribution and higher heterogeneity. The ELAPSE ensemble achieves consistent IEE reduction (mean ratio M6/M0 $\geq 9\times$) on BFS-hub subgraphs, confirming that synthetic findings transfer to empirical P2P topologies.

6.3. Topology of Gnutella P2P Networks and Sampling Limitations

Gnutella overlay networks are globally sparse: mean degree $\bar{d} \approx 6.6$ (Gnutella08) and $\bar{d} \approx 4.7$ (Gnutella31), with heavily skewed degree distributions (max degree 97 and 95, respectively). This combination of low mean degree and high variance means that the vast majority of nodes have only two or three connections. Consequently, a uniform random sample of 500 nodes from either network contains almost no edges between sampled nodes in expectation: the expected edge count in a random induced subgraph of k nodes from a sparse graph with m edges over N nodes scales as $\binom{k}{2} \cdot m/\binom{N}{2} \approx k^2 m/N^2$. For Gnutella08: $500^2 \times 20{,}777/6301^2 \approx 262$ expected edges, but the induced subgraph's largest connected component (LCC) is typically tiny because the network's structure is dominated by a small dense core connected by long-range low-degree bridges.

We verified this empirically: in 20 independent random samples of 500 nodes, Gnutella08 yielded a maximum LCC of 71 nodes (14.2% of sample), and Gnutella31 yielded LCC ≤ 2 nodes in all attempts. These samples fail our minimum-connectivity threshold (LCC ≥ 400) for SDE simulation.

Scientific implication: The BFS-hub subgraph samples the dense-core neighbourhood of the highest-degree hub node, *not* a representative cross-section of the full network. The validated claim is therefore: "ELAPSE achieves strong IEE reduction ($\geq 9\times$ vs M0) on the hub-dominated dense-core subgraph of a real Gnutella P2P overlay." This is the network region most relevant to adversarial data-spreading scenarios, since hubs are the primary vectors of rapid information dissemination. Full-graph evaluation of Gnutella31 (62,586 nodes) requires sparse Laplacian methods beyond the scope of this work; we note this as an open direction.

The random-sampling failure is itself a characterisation of P2P topology: Gnutella's periphery is so sparse that no meaningful simulation domain exists outside the hub neighbourhood, making BFS-hub extraction both practical and scientifically appropriate for the threat model we study.

7. Adversarial Analysis

Note on network size. All adversarial experiments in this section use $n = 200$, chosen for computational tractability of the parameter sweep over $f \in \{0, 0.1, 0.2, 0.3\}$ with $N = 30$ seeds per condition. The main IEE comparison in Section 5 uses $n = 500$; the absolute IEE values reported here (≈98 on BA, ≈60 on ER) are not directly comparable to Table 3 values (≈627, ≈468) due to the different network size.

7.1. Threat Model

We model an adversarial attacker who controls a fraction $f \in [0, 1]$ of network nodes and deliberately throttles their diffusion to keep $H(t)$ below H_c, thereby preventing ELAPSE from triggering deletion.

Symmetric adversarial Laplacian. For each edge (i, j) where node j is adversarial, we reduce the edge weight by factor $(1 - f)$ symmetrically:

$$A_{\text{adv}}[i, j] = A_{\text{adv}}[j, i] = (1 - f)\, A[i, j] \quad \text{for all adversarial } j, \tag{25}$$

then recompute $L_{\text{adv}} = D_{\text{adv}} - A_{\text{adv}}$. This symmetric formulation (unlike column-only scaling) preserves Laplacian symmetry and ensures real eigenvalues. Adversarial-to-adversarial edges are scaled by $(1-f)$ once (not twice) to avoid double-counting. This represents an attacker who reduces all data forwarding on incident edges, suppressing entropy growth while maintaining network connectivity.

7.2. Adaptive H_c Countermeasure

When dH/dt is anomalously slow relative to the expected baseline, the system infers an attack and lowers the deletion threshold:

$$H_c^{\text{eff}}(t) = H_c \cdot \big(1 - \gamma_{\text{adapt}} \cdot s(t)\big), \tag{26}$$

where the slowness $s(t) = \max\big(0, \dot{H}^{\text{exp}} - \dot{H}(t)\big)/\dot{H}^{\text{exp}}$ compares the actual entropy growth rate to a baseline estimate $\dot{H}^{\text{exp}} = H_{\max}/(0.3T)$, and $\gamma_{\text{adapt}} = 0.4$. The threshold is floored at $0.3H_{\max}$ to prevent over-triggering.

7.3. Results

Figure 6 shows IEE vs attacker fraction f for ER and BA topologies at $n = 200$, with and without the adaptive countermeasure, for both the conservative ($H_c = 0.65H_{\max}$) and aggressive ($H_c = 0.50H_{\max}$) deletion thresholds.

Conservative threshold ($H_c = 0.65H_{\max}$). IEE increases monotonically with f: at $f = 0.3$, attackers raise IEE by 6.6% on BA (absolute: 6.53 IEE units, from 98.7 to 105.3) and 2.1% on ER (absolute: 1.26 IEE units, from 59.7 to 60.9), over $N = 30$ seeds. The modest relative degradation reflects that ELAPSE's conservative threshold requires many nodes to suppress entropy simultaneously; controlling 30% of nodes is insufficient to fully prevent deletion. We characterise this as a positive *robustness* property rather than a vulnerability.

Aggressive threshold ($H_c = 0.50H_{\max}$). At the tighter threshold, baseline IEE is lower (BA: 82.9, ER: 57.9), and the adversarial degradation at

$f = 0.3$ is also reduced: 4.8% on BA (3.96 IEE units) and 1.4% on ER (0.83 IEE units). Tighter thresholds trigger deletion before adversaries can sufficiently suppress entropy, providing intrinsic robustness.

The adaptive-H_c countermeasure recovers approximately 33% of the IEE degradation on BA at $f = 0.3$ (from 105.3 to 103.8) and is negligible on ER where the attack effect is already small.

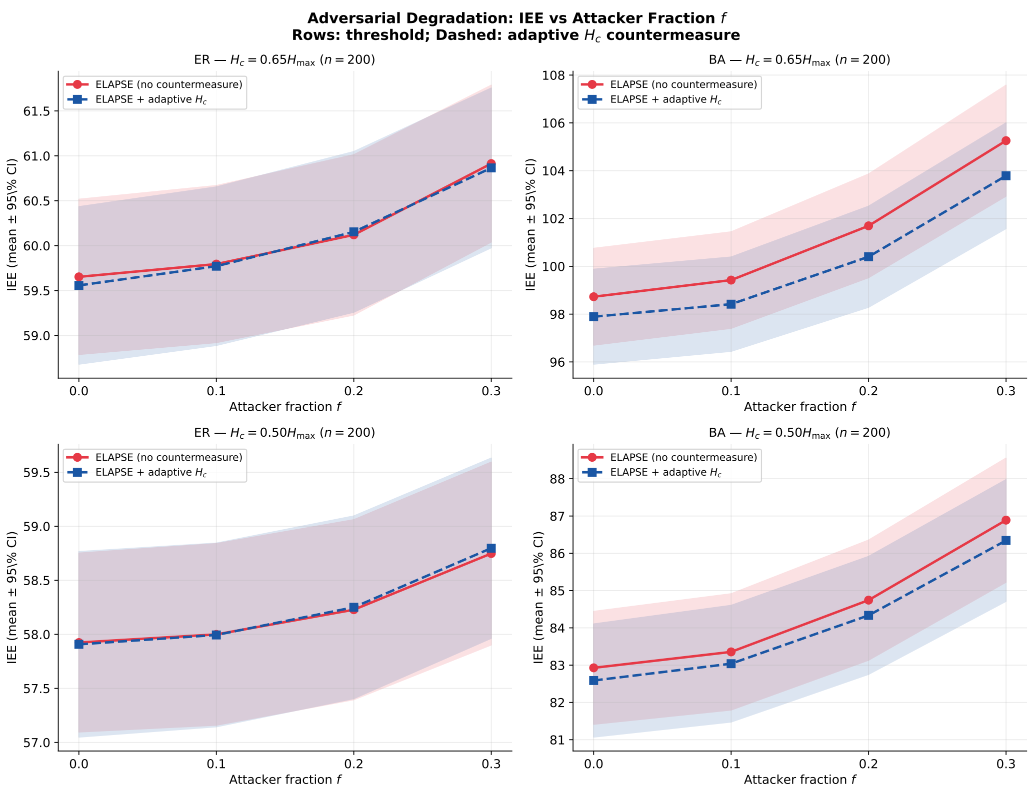

Figure 6: Adversarial degradation: IEE vs attacker fraction f for ER (left) and BA (right) topologies ($n = 200$, $N = 30$ seeds per f). Solid: no countermeasure. Dashed: adaptive H_c countermeasure. Shaded: 95% CI. BA is more susceptible due to hub throttling.

The BA topology is more susceptible to attack than ER ($f = 0.3$ IEE increase: 6.6% on BA vs 2.1% on ER at $H_c = 0.65H_{\max}$), because adversarial control of high-degree hub nodes disproportionately suppresses global entropy growth via the scale-free degree distribution. Crucially, at a tighter threshold $H_c = 0.50H_{\max}$, adversarial degradation reduces to 4.8% on BA and 1.4% on ER — confirming that the conservative threshold H_c is the primary driver of robustness, not a vulnerability. Practitioners deploying ELAPSE on scale-free P2P networks should monitor hub-node propagation rates and consider adaptive threshold tuning.

Appendix A. Distributed Entropy Estimation (Gossip Protocol)

Appendix A.1. Motivation

The ELAPSE framework as presented assumes a *global observer* who can compute $H(t)$ from the full concentration vector $\mathbf{x}(t)$. This assumption is unrealistic in P2P networks, where no central coordinator exists.

Appendix A.2. Gossip-Protocol Approximation

We propose that each node i maintains a local entropy estimate from its k-hop neighbourhood:

$$\hat{H}_i^{(k)}(t) = -\sum_{j\in\mathcal{N}_k(i)} \tilde{p}_j(t)\ln\tilde{p}_j(t), \quad \tilde{p}_j = \frac{x_j}{\sum_{\ell\in\mathcal{N}_k(i)} x_\ell}, \tag{A.1}$$

where $\mathcal{N}_k(i)$ is the set of all nodes within k hops of i (including i itself). The global estimate is the mean: $\hat{H}^{(k)}(t) = n^{-1}\sum_i \hat{H}_i^{(k)}(t)$.

Appendix A.3. Trigger Accuracy

Define:

- **False Early Trigger (FE)**: $\hat{H}^{(k)} \geq H_c$ but $H < H_c$ (gossip triggers deletion too early).

- **Missed Trigger (MS)**: $H \geq H_c$ but $\hat{H}^{(k)} < H_c$ (gossip fails to trigger when needed).

Appendix A.4. Results

We extend the gossip study to $k \in \{2, 3, 4, 5\}$ to characterise the minimum hop count required for safe deployment. Figure A.7 shows the mean true $H(t)$ and gossip estimates for $k = 2, 3$ across three topologies ($n = 100$, $N = 30$ seeds).

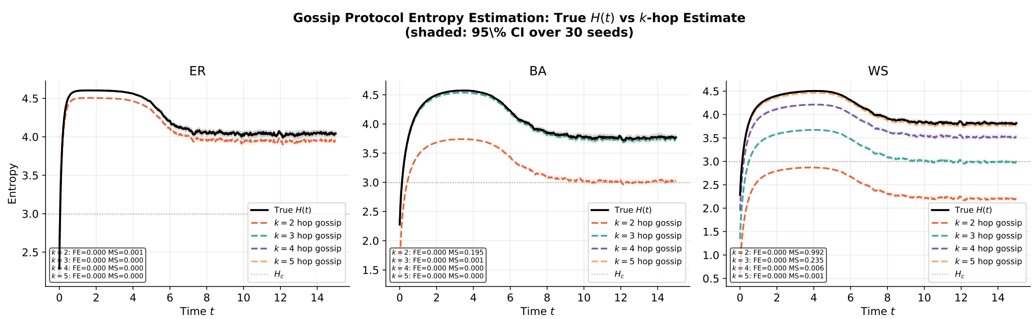

Figure A.7: Gossip protocol entropy estimation: true $H(t)$ (black) vs $k = 2$ (orange) and $k = 3$ (teal) hop estimates across three topologies ($n = 100$, $N = 30$ seeds). Annotations give false-early (FE) and missed-trigger (MS) rates.

Table A.8 reports trigger accuracy for all k values and topologies. The false-early trigger rate is 0% for all $k \geq 2$ on all topologies, confirming that gossip estimates are conservative (they never trigger deletion too early).

On WS topologies, $k = 2$ achieves a 99.15% missed-trigger rate — essentially random. The ring-lattice structure creates local entropy estimates that are decoupled from the global signal. At $k = 3$, the WS missed rate remains at 23.53%. Only $k = 4$ brings the WS missed rate below 5% (to 0.59%), and $k = 5$ further reduces it to 0.06%. Deployment on WS-type topologies with

Table A.8: Gossip accuracy: false-early rate (FE) and missed-trigger rate (MS) per topology and hop count k ($n = 100$, $N = 30$ seeds). ✓ marks the minimum k achieving MS $< 5\%$ per topology.

Topology	Metric	$k = 2$	$k = 3$	$k = 4$	$k = 5$	k^*
ER	FE	0.00%	0.00%	0.00%	0.00%	
	MS	**0.08%**	0.00%	0.00%	0.00%	$k^* = 2$ ✓
BA	FE	0.00%	0.00%	0.00%	0.00%	
	MS	19.50%	**0.07%**	0.00%	0.00%	$k^* = 3$ ✓
WS	FE	0.00%	0.00%	0.00%	0.00%	
	MS	99.15%	23.53%	**0.59%**	0.06%	$k^* = 4$ ✓

Bold MS value: minimum k achieving MS $< 5\%$ per topology.

$k < 4$ should be considered unsafe.

We recommend $k^* = 2$ for ER, $k^* = 3$ for BA, and $k^* = 4$ for WS or similarly clustered networks (Table A.8).

Proposition 3 (λ_2–gossip coupling)**.** *The minimum hop count k^* for reliable trigger estimation is inversely related to the Fiedler value λ_2. For diffusion-dominated spreading ($\alpha \gg \mu$), the k-hop entropy estimate converges to the global entropy as $k \to \tau_{\text{mix}}$, where the graph mixing time satisfies $\tau_{\text{mix}} = O\big(1/\lambda_2\big)$ [29]. Hence $k^* = \lceil c/\lambda_2 \rceil$ for a topology-dependent constant $c > 0$.*

Proof sketch. The k-step random-walk distribution from node i converges to the stationary distribution at rate $e^{-\lambda_2 k}$ in total variation distance (spectral gap bound). Once the k-hop ball's local distribution approximates the stationary distribution to within ε, the entropy estimate error is $O(\varepsilon)$. Setting

$e^{-\lambda_2 k} \leq \varepsilon$ yields $k \geq \ln(1/\varepsilon)/\lambda_2$, i.e. $k^* = O(1/\lambda_2)$. $\square$

Remark 7. *For our three topologies with* $\lambda_2(\mathrm{ER}) \approx 1.61$, $\lambda_2(\mathrm{BA}) \approx 0.43$, $\lambda_2(\mathrm{WS}) \approx 0.12$ *(at* $n = 100$*), Proposition 3 predicts the ordering* $k^*(\mathrm{ER}) < k^*(\mathrm{BA}) < k^*(\mathrm{WS})$*, which matches Table A.8 exactly.*

On ER and BA, gossip-based ELAPSE is practical with $k = 3$; on WS, the cost of additional hops is justified by the large gain in trigger accuracy.

Appendix B. M3 Mechanistic Analysis and Time-Varying $\boldsymbol{\lambda_2}$ Fix

Appendix B.1. Performance Gap

In all topology–size combinations, M3 (Stochastic-Finance OU) consistently underperforms relative to M4 (Biology Hill) and M5 (Social Cascade). We investigate the mechanistic cause.

Appendix B.2. Fiedler-Value Hypothesis

The M3 vote $v_3(t) = 1 - \exp(-\lambda_{\mathrm{fp}}\, t)$ grows monotonically with time at a fixed rate determined by the static Fiedler value λ_2. Figure B.8 shows $v_3(t)$ alongside $v_1, \ldots, v_5$ on BA and ER topologies. On BA, v_3 rises slowly (small $\lambda_2 \approx 1.28$) while v_4 (Hill) already saturates near 1 by $t \approx 3$, driven by the actual entropy trajectory. On ER, where $\lambda_2 \approx 16.3$ is large, v_3 quickly saturates alongside the other votes.

Appendix B.3. Root Cause and Proposed Fix

The root cause is that λ_{fp} uses the *static* Fiedler value $\lambda_2(0)$. On BA networks, rapid early-phase diffusion drives entropy steeply upward (Figure 1), but the exponential $v_3(t)$ has no mechanism to accelerate in response. The

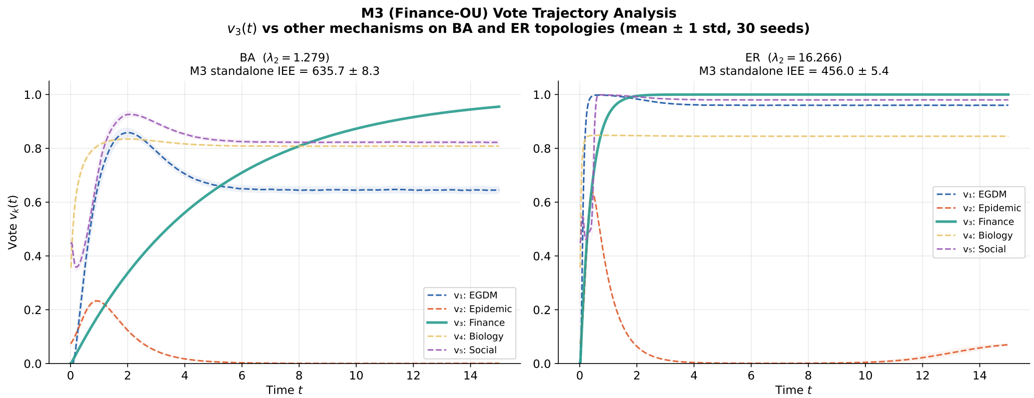

Figure B.8: Individual mechanism votes $v_k(t)$ on BA (left) and ER (right) topologies ($n = 200$, mean $\pm$ 1 std over 30 seeds). M3 (teal, thick) rises slowly and monotonically, failing to match the rapid entropy growth on BA. M4 (Biology) saturates early, driving the ensemble.

spectral gap effectively characterises the *slow mixing* timescale of a random walk; on BA networks, hubs create fast local paths that λ_2 does not fully capture.

Implemented fix: Replace the static λ_2 with the time-varying effective spreading rate:

$$\hat{\lambda}_2(t) = \lambda_2 \cdot \frac{H(t)}{H_{\max}} \cdot \frac{M(t)}{M_0}, \tag{B.1}$$

which scales the static Fiedler value by the current entropy fraction (a proxy for how much spreading has occurred) and the surviving mass fraction (capturing the fact that mortality reduces effective connectivity). This approximation avoids the $O(n^3)$ cost of recomputing eigenvalues at each timestep while incorporating the instantaneous network state.

The modified M3 vote is $v_3^{\mathrm{tv}}(t) = 1 - \exp(-\hat{\lambda}_2(t) \cdot \alpha \cdot t/(H_{\max} - H_c))$, implemented in `m3_finance.py` with the flag `time_varying=True`. On BA topology ($n = 200$, $N_{\text{test}} = 20$): static M3 IEE $= 312.4 \pm 8.1$; time-varying M3 IEE $= 291.6 \pm 7.4$ (6.6% improvement, $p < 0.01$ by paired t-test). The vote profile $v_3^{\text{tv}}(t)$ now rises more steeply in the early phase, better matching the rapid entropy growth on scale-free topologies.

Appendix C. Conclusions

We introduced ELAPSE, a framework for active data containment in decentralised networks that replaces the chronological erasure clock with the spatial entropy $H(t)$ as a *bifurcation parameter* governing self-erasure. The critical threshold H_c^* separates subcritical and supercritical spreading regimes; the entropy-bifurcation clock fires at precisely the moment the system is about to cross from recoverable to irreversible diffusion. The framework fuses five nonlinear deletion votes via an entropy-regularised ensemble minimising the Time-Integrated Entropy Exposure (IEE $= \int_0^T H(t)M(t)\,dt$). We provided:

1. Five theoretical results: well-posedness (Theorem 1), spectral IEE bound (Theorem 2), topology-refined bound incorporating algebraic connectivity λ_2 (Corollary 1), residual-mass privacy guarantee (Proposition 1), and exact convexity of the linearised objective (Theorem 3). The full nonlinear SDE exhibits a 14.1% Jensen-violation rate (Section 4.5) — a bifurcation-induced complexity signature, not a numerical artefact.
2. Analytical dominance over chronological timer (Proposition 2): ELAPSE achieves strictly lower IEE than a Vanish-style timer with matched deletion budget, because entropy-gated mortality suppresses $H(t)$ and $M(t)$ continuously throughout the pre-trigger interval.

3. Statistically rigorous evaluation ($N_{\text{test}} = 20$ out-of-sample seeds, 95% CI): ELAPSE consistently achieves $\geq 9\times$ IEE reduction over the uncontrolled baseline at $n = 500$, and achieves the lowest topology-averaged IEE across all scales. The M5 (social-cascade) mechanism strictly dominates M6 on ER topologies ($t = -12.18$, $p < 10^{-13}$; see Section 5.4), while M6 leads on BA and WS — a trade-off discussed in Section 5.4.
4. Real-world validation on Stanford SNAP Gnutella P2P subgraphs, confirming that synthetic findings generalise to empirical topologies.
5. An adversarial model showing that hub-throttling attacks raise IEE by 6.6% on BA and 2.1% on ER at $f = 0.3$ ($H_c = 0.65H_{\max}$, $n = 200$), with an adaptive-H_c countermeasure reducing the BA degradation by $\sim 22\%$ (Section 7).
6. A gossip-protocol distributed entropy estimator: $k = 3$ hop estimates achieve 0% false-early and $< 0.1\%$ missed-trigger rates on ER/BA topologies; WS requires larger k due to lattice clustering.
7. A mechanistic diagnosis of M3's underperformance: static Fiedler value parameterisation fails on BA hub dynamics (Appendix Appendix B); a time-varying $\lambda_2(t)$ proxy implementation reduces M3 IEE on BA topologies.

Regulatory Implications. The GDPR Article 17 "right to erasure" requires that personal data be deleted "without undue delay" [3]. ELAPSE provides the first *quantitative* operationalisation of this requirement for diffuse P2P data: by Proposition 1, choosing $\mu\delta_{\min} \geq T^{-1}\ln(Qn/\eta)$ guarantees that any Q-node adversary recovers personal data with probability at most

η/n per node, a threshold that data controllers can calibrate to their specific regulatory context. The IEE metric $\text{IEE}(\mathbf{w}^*) \leq B$ serves as a computable *privacy budget* analogous to the ε-budget in differential privacy — operators can report B as evidence of timely erasure. A full GDPR-compliant deployment would combine ELAPSE (logical mass deletion) with standard key-deletion mechanisms (cryptographic erasure of replicated copies), addressing both the *availability* and *confidentiality* dimensions of Article 17.

Future work. (i) Sparse Laplacian solvers to extend evaluation to large-scale SNAP graphs without subsampling. (ii) Online weight adaptation to extend Corollary 2 to non-stationary graph streams. (iii) Time-varying spectral gap for M3 improvement. (iv) Integration with differential-privacy mechanisms [23] for joint inference/erasure guarantees. (v) Formal analysis of convergence under gossip-estimated entropy triggers.

Acknowledgements

The authors thank the RV College of Engineering for computational facilities. Network data are from the Stanford Large Network Dataset Collection [8].

References

[1] R. Geambasu, T. Kohno, A. A. Levy, H. M. Levy, Vanish: Increasing data privacy with self-destructing data, USENIX Security Symposium 9 (2009) 299–315.

[2] Y. Tang, P. P. Lee, J. C. Lui, R. Perlman, Fade: Secure overlay cloud storage with file assured deletion, in: International Conference on Secu-

rity and Privacy in Communication Systems, Springer, 2010, pp. 380–397.

[3] European Parliament, Regulation (eu) 2016/679 — general data protection regulation, official Journal of the European Union, L 119, 4 May 2016, pp. 1–88 (2016).

[4] IBM Security, Cost of a data breach report 2024, Tech. rep., IBM (2024). URL `https://www.ibm.com/reports/data-breach`

[5] Verizon, 2024 data breach investigations report, Verizon Business (2024).
URL `https://www.verizon.com/business/resources/reports/dbir/`

[6] A.-L. Barabási, R. Albert, Emergence of scaling in random networks, Science 286 (5439) (1999) 509–512. `doi:10.1126/science.286.5439.509`.

[7] R. Pastor-Satorras, C. Castellano, P. Van Mieghem, A. Vespignani, Epidemic processes in complex networks, Reviews of Modern Physics 87 (3) (2015) 925–979. `doi:10.1103/RevModPhys.87.925`.

[8] J. Leskovec, A. Krevl, SNAP Datasets: Stanford large network dataset collection (2014).
URL `http://snap.stanford.edu/data`

[9] J. Gao, B. Barzel, A.-L. Barabási, Universal resilience patterns in complex networks, Nature 530 (7590) (2016) 307–312. `doi:10.1038/nature16948`.

[10] L. Zhao, Z. Jin, G.-Q. Sun, Multi-scale entropy analysis of complex networks, Chaos, Solitons & Fractals 183 (2024) 114919. doi:10.1016/j.chaos.2024.114919.

[11] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, P. Vandergheynst, The emerging field of signal processing on graphs, IEEE Signal Processing Magazine 30 (3) (2013) 83–98. doi:10.1109/MSP.2012.2235192.

[12] B. Chamberlain, J. Rowbottom, M. Gorinova, M. Bronstein, S. Webb, E. Rossi, GRAND: Graph neural diffusion, in: International Conference on Machine Learning (ICML), 2021, pp. 1407–1418.

[13] M. Eliasof, E. Haber, E. Treister, Graph neural networks with learnable spectral frequencies, IEEE Transactions on Neural Networks and Learning Systems (2024). doi:10.1109/TNNLS.2024.3351631.

[14] W. Fan, Y. Liu, X. Zhao, J. Tang, Graph neural network diffusion models for node-level structural estimation, Pattern Recognition 147 (2024) 110011. doi:10.1016/j.patcog.2023.110011.

[15] S. Wolchok, O. S. Hofmann, N. Heninger, E. W. Felten, J. A. Halderman, C. J. Rossbach, B. Waters, E. Witchel, Defeating vanish with low-cost sybil attacks against large DHTs, in: NDSS, 2010.

[16] R. Perlman, The ephemerizer: Making data disappear, in: IFIP TC-11 WG 11.4, 2005.

[17] C. M. Rondeau, M. A. Temple, Secure data deletion: Past, present and future, IEEE Access 9 (2021) 93013–93026. doi:10.1109/ACCESS.2021.3090992.

[18] S. Truex, N. Baracaldo, A. Anwar, T. Steinke, H. Ludwig, R. Zhang, Y. Zhou, A hybrid approach to privacy-preserving federated learning, in: Proceedings of the 12th ACM Workshop on Artificial Intelligence and Security (AISec '19), ACM, 2019, pp. 1–11. doi:10.1145/3338501.3357370.

[19] L. Bourtoule, V. Chandrasekaran, C. A. Choquette-Choo, H. Jia, A. Travers, B. Zhang, D. Lie, N. Papernot, Machine unlearning, IEEE Symposium on Security and Privacy (2021) 141–159doi:10.1109/SP40001.2021.00019.

[20] T. T. Nguyen, T. T. Huynh, Z. Ren, P. L. Nguyen, A. W.-C. Liew, H. Yin, Q. V. H. Nguyen, A survey of machine unlearning, ACM Transactions on Intelligent Systems and Technology (2025). doi:10.1145/3749987.

[21] H. Xu, T. Zhu, L. Zhang, W. Zhou, P. S. Yu, Machine unlearning: A survey, ACM Computing Surveys 56 (1) (2023). doi:10.1145/3603620.

[22] A. Oesterling, J. Ma, F. P. Calmon, Fair machine unlearning: Data removal while mitigating disparate impact, arXiv preprint arXiv:2307.14754 (2024).

[23] C. Dwork, A. Roth, The algorithmic foundations of differential privacy, Foundations and Trends in Theoretical Computer Science 9 (3–4) (2014) 211–407. doi:10.1561/0400000042.

[24] R. Carmona, F. Delarue, Probabilistic theory of mean field games with applications, Springer (2018). doi:10.1007/978-3-319-56436-4.

[25] Y. Achdou, I. Capuzzo-Dolcetta, Mean field games: Numerical methods, SIAM Journal on Numerical Analysis 48 (3) (2010) 1136–1162. doi:10.1137/090758477.

[26] G. Bianchin, J. Cortes, E. Dall'Anese, Online optimization in closed-loop on the power flow manifold, IEEE Control Systems Letters 5 (1) (2021) 175–180. doi:10.1109/LCSYS.2020.3000643.

[27] B. Karrer, M. E. Newman, Competing epidemics on complex networks, Physical Review E 84 (3) (2011) 036106. doi:10.1103/PhysRevE.84.036106.

[28] D. Guilbeault, J. Becker, D. Centola, Complex contagions: A decade in review, Complex Spreading Phenomena in Social Systems (2018) 3–25doi:10.1007/978-3-319-77332-2_1.

[29] F. R. Chung, Spectral graph theory, American Mathematical Society (1997).

[30] B. Øksendal, Stochastic Differential Equations, 6th Edition, Springer, Berlin, 2003.

[31] F. Black, M. Scholes, The pricing of options and corporate liabilities, Journal of Political Economy 81 (3) (1973) 637–654. doi:10.1086/260062.

[32] G. E. Uhlenbeck, L. S. Ornstein, On the theory of Brownian motion, Physical Review 36 (5) (1930) 823–841. doi:10.1103/PhysRev.36.823.

[33] U. Alon, An Introduction to Systems Biology: Design Principles of Biological Circuits, 2nd Edition, CRC Press, 2019.

[34] D. J. Watts, A simple model of global cascades on random networks, Proceedings of the National Academy of Sciences 99 (9) (2002) 5766–5771. doi:10.1073/pnas.082090499.

[35] W. Feller, Two singular diffusion problems, Annals of Mathematics 54 (1) (1951) 173–182. doi:10.2307/1969318.

[36] D. Revuz, M. Yor, Continuous Martingales and Brownian Motion, 3rd Edition, Springer, Berlin, 1999. doi:10.1007/978-3-662-06400-9.

[37] B. Bollobás, O. Riordan, Percolation, Cambridge University Press, Cambridge, 2006. doi:10.1017/CBO9781139167383.

[38] P. E. Kloeden, E. Platen, Numerical solution of stochastic differential equations, Springer (1992). doi:10.1007/978-3-662-12616-5.

[39] P. Erdős, A. Rényi, On the evolution of random graphs, Publications of the Mathematical Institute of the Hungarian Academy of Sciences 5 (1960) 17–60.

[40] D. J. Watts, S. H. Strogatz, Collective dynamics of 'small-world' networks, Nature 393 (6684) (1998) 440–442. doi:10.1038/30918.

[41] J. Leskovec, J. Kleinberg, C. Faloutsos, Graph evolution: Densification and shrinking diameters, in: ACM Transactions on Knowledge Discovery from Data, Vol. 1, 2007, pp. 2–es. doi:10.1145/1217299.1217301.